PRODUCT USER MANUAL

For Wind- Global Ocean L3 Wind

WIND_GLO_PHY_L3_NRT_012_002

and

WIND_GLO_PHY_L3_MY_012_005

Issue: 2.27

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0 2.0 2.1 2.2 2.3	15 Sept. 2011 28 Sept. 2012 14 Feb. 2013 10 Feb 2015	All	Creation of the document Changes are template change Extended description of product content New datasets added U10S, curl, divergence	Tilly Driesenaar Tilly Driesenaar Tilly Driesenaar Ad Stoffelen, Jos de Kloe	Ad Stoffelen Ad Stoffelen Ad Stoffelen Ad Stoffelen Lars-Anders Breivik L. Crosnier
2.4	20 May 2015	all	Change format to fit Copernicus Marine graphical rules		L. Crosnier
2.5	22 Sept. 2015	all	Finish Copernicus Marine rebranding	Jos de Kloe, Bruce Hackett	
2.6	19 Jan 2016		Product change	Jos de Kloe	
2.7	3 Feb. 2016		Revert name change for scat measurements and add standard names	Jos de Kloe	
2.8	12 Jul. 2016		NetCDF format changes and addition of wind curl and divergence fields	Jos de Kloe	B. Hackett
2.9	13 Jul 2016		Minor updates on MOTU and FTP URL/filenames examples	V Forneris	B. Hackett
2.10	6 Dec. 2016	V.2.2	Correct for scaling bug for curl/div in NetCDF description	Jos de Kloe	
2.11	6 Jan. 2017	I, III.1, III.2, IV.1, V.2,	Remove reference to V1 file format for ASCAT products. and add REP product	Jos de Kloe	
2.12	5 Sep. 2017		Extended ASCAT-A REP time-series and added ERS-1/ERS-2 REP data	M. Belmonte	
2.13	26 Jan 2018		Rebranding from OSI to Wind TAC	M. Belmonte	
2.14	21 Aug 2018		Added OSCAT2 REP data, ScatSat-1 NRT data, and time-series extension of ASCAT-A REP	M. Belmonte	
2.15	9 Jan 2019		Time-series extension of ASCAT-A REP	M. Belmonte	
2.16	11 April 2019		Time-series extension of ASCAT-A REP	M. Belmonte GJ van Zadelhoff	

2.17	4-Sep-2019		Added Metop-C ASCAT data, NetCDF-4 format, expanded acronym list	R. Giesen	
2.18	25-Mar-2020		Change from ERA-Interim to ERA5 model wind collocation and file format change for REP time-series, time series extension of ASCAT-A REP	R. Giesen	
2.19	10-Sep-2020		Added HY-2B HSCAT NRT datasets and information about REP regular time series extension. Updated Table 3 conform the PUM template.	R. Giesen	C. Derval
2.20	16-Jul-2021		Discontinuation of ScatSat-1 NRT datasets	R. Giesen	C. Derval
2.21	10-Sep-2021		Added HY-2C HSCAT NRT datasets and Metop-B ASCAT REP datasets	R. Giesen	C. Derval
2.22	16-Nov-2021		Discontinuation of Metop-A ASCAT NRT datasets	R. Giesen	C. Derval
2.23	31-Mar-2022		Changed update frequency of Metop-B ASCAT REP datasets to monthly. Metop-A ASCAT REP datasets discontinued.	R. Giesen	C. Derval
2.24	13-Jun-2022		Added HY-2D HSCAT NRT datasets	R. Giesen	Copernicus Marine Product Management
2.25	13-Jun-2023	All	Changed product and dataset names, inclusion of winds closer to the coast for 0.125° datasets, update of ERS-1, ERS-2 and QuikSCAT, Oceansat-2 MY datasets	R. Giesen	Copernicus Marine Product Management
2.26	30-Jan-2024		Added Oceansat-3 OSCAT NRT datasets	R. Giesen	
2.27	27-May-2025		Updated for reprocessed Metop-A ASCAT MY datasets	R. Giesen	Copernicus Marine Product Management

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GLOSSARY AND ABBREVIATIONS

BUFR	Binary Universal Form for the Representation of meteorological data
CF	Climate Forecast (convention for netCDF)
CMOD	C-band geophysical model function
CNR	Consiglio Nazionale delle Ricerche
COARE	Coupled Ocean-Atmosphere Response Experiment
ECMWF	European Centre for Medium-Range Weather Forecasts
ERA	ECMWF Re-analysis
KNMI	Royal Netherlands Meteorological Institute
IFREMER	Institut Français pour la Recherche et l'Exploitation de la MER
ISRO	Indian Space Research Organisation
netCDF	Network Common Data Form
NRT	Near Real-Time
NWP	Numerical Weather Prediction
OSI SAF	Ocean and Sea Ice Satellite Application Facility
PC	Production Center
PU	Production Unit
REP	Reprocessed
U10S	10m-height stress-equivalent winds
Wind Meridional Component	West to East component of wind-to vector
Wind Zonal Component	South to North component of the wind-to vector

WVC	Wind Vector Cell

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

I INTRODUCTION

I.1 Summary

This guide describes data product files from the Copernicus Marine Production Unit at the Royal Netherlands Meteorological Institute (KNMI), what data services are available to access them, and how to use the files and services.

The **WIND_GLO_PHY_L3_NRT_012_002** product contains near real-time (NRT) L3 gridded sea surface wind observations from scatterometer. The product is produced by KNMI based on the Near-Real-Time L2 scatterometer product and distributed by the Copernicus Marine Service.

The **WIND_GLO_PHY_L3_MY_012_005** product contains multiyear (MY) gridded sea surface wind observations from scatterometer. This product is produced by KNMI based on the reprocessed L2 scatterometer products and distributed by the Copernicus Marine Service.

These Copernicus Marine products contain L3 global daily gridded scatterometer observations relying on upstream L2 scatterometer products from the EUMETSAT Ocean and Sea Ice Satellite Application Facility (OSI SAF) Wind Centre at KNMI. The different datasets within the product may have different grid spacings, varying from 12.5 km and 25 km (Metop-A, B and C), 25 km (ERS-1 and ERS-2) to 25 km and 50 km (QuikSCAT, ScatSat-1, Oceansat-2, Oceansat-3, HY-2B, HY-2C and HY-2D), depending on the scatterometer and upstream processing. Data from ascending and descending passes are gridded into separate files. OSI SAF L2 quality monitoring flags are transferred to the L3 grid and included in the Copernicus Marine L3 products.

The L3 MY time series are based on the L2 climate data records (CDR) reprocessed within OSI SAF. The data records for ERS-1, ERS-2, QuikSCAT Oceansat-2 and Metop-A have recently been reprocessed, using European Centre for Medium-range Weather Forecasts (ECMWF) ERA5 reanalysis model winds to remove the ambiguity in the wind direction. Until newly reprocessed datasets for Metop-B become available (expected in 2026), the Metop-B datasets use ERA-Interim reanalysis data for the ambiguity removal. However, ECMWF stopped the production of ERA-Interim data in August 2019, therefore L2 reprocessing with ERA-Interim is not possible for more recent dates. Hence, L2 NRT data has been used to update the L3 MY time series, starting from January 2019. Since the L2 reprocessed and NRT datasets are processed with the same processing software version, the only difference is the source of the ECMWF model wind (ECMWF ERA-Interim and operational model, respectively) used in the scatterometer processing to remove the ambiguity in the wind direction. The effect of this difference on the continuity of the L3 MY time series is expected to be negligible.

The Copernicus Marine L3 global wind product also contains gridded model winds from ECMWF. The NRT products use the ECMWF Operational numerical weather predication (NWP)

model, while the MY products use the ECMWF ERA5 reanalyses. The model winds are collocated in space and time with the scatterometer observations in the OSI SAF upstream L2 wind product. The model wind in the Copernicus Marine L3 global wind product is then sampled and processed in exactly the same way as the scatterometer winds, so subject to identical space and time sampling errors. Comparing scatterometer-sampled ECMWF winds with uniformly-sampled ECMWF winds over a period of interest, reveals these spatio-temporal scatterometer sampling errors.

In 2016, a significant file format change has been implemented in the L3 production, including the addition of wind stress, curl and divergence of wind and wind stress, stress equivalent reference winds, the associated air density field, and some more technical changes in the definition of dimensions and attributes. The file format change for ASCAT has been applied to the full period of NRT and MY scatterometer data available in the Copernicus Marine archives. This change is the result of progress in understanding how the scatterometer measurements should be used (stress equivalent reference winds), in addition to user required fields (stress, rotation, divergence). It is used for all new near real-time and reprocessed data sets.

Following updates in the Level-1 products from 22 November 2022 onwards, the minimum distance to the coast in the L2 OSI SAF Metop ASCAT coastal wind products was reduced from approximately 20 km to 5-10 km. As a consequence, the L3 NRT datasets for Metop-B and Metop-C ASCAT at 0.125° contain information for winds and derived variables closer to the coast from this date onwards. The newly reprocessed L3 MY coastal datasets for Metop-A ASCAT (released in November 2025) also contain winds closer to the coast. The L3 MY datasets for Metop-B ASCAT have not yet been reprocessed, but are based on the L3 NRT winds from 2019 onwards.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

The different instruments available in this product have been updated for different Copernicus Marine releases. Details are given in the following table (Table 1).

Product Specification	WIND_GLO_PHY_L3_NRT_012_002 and WIND_GLO_PHY_L3_MY_012_005
Geographical coverage	0°E → 360°E ; 90°S → 90°N
Temporal resolution	Daily files with instantaneous measurements
Target delivery time	Daily (WIND_GLO_PHY_L3_NRT_012_002) Monthly (WIND_GLO_PHY_L3_MY_012_005)
Delivery mechanism	Marine Information System: SUBSETTER, FTP
Number of vertical levels	1 at 10 meters
Analysis or processing	Processing
Forecast	No
Data Source	ASCAT on Metop-A ASCAT on Metop-B ASCAT on Metop-C OSCAT on Oceansat-2 OSCAT on Oceansat-3 OSCAT on ScatSat-1 HSCAT on HY-2B HSCAT on HY-2C HSCAT on HY-2D SeaWinds on QuikSCAT SCAT on ERS-1 SCAT on ERS-2
Format	NetCDF-3 CF-1.6 (Metop-A, B and ScatSat-1 NRT time series)

	NetCDF-4 classic model CF-1.6 (all MY time series and Metop-C, Oceansat-3, HY-2B, HY-2C and HY-2D NRT)
Horizontal resolution	0.25 degrees and 0.125 degrees for Metop-A, B and C 0.5 degrees and 0.25 degrees for QuikSCAT, Oceansat-2, Oceansat-3, ScatSat-1, HY-2B, HY-2C and HY-2D 0.25 degrees for ERS-1 and ERS-2
Available NRT time series	From 1-Jan-2016 to 15-Nov-2021 (Metop-A) From 1-Jan-2016 to present day (Metop-B) From 28-Oct-2019 to present day (Metop-C) From 21-Oct-2018 to 28-Feb-2021 (ScatSat-1) From 11-Nov-2020 to present day (HY-2B) From 8-Nov-2021 to present day (HY-2C) From 15-Oct-2022 to 3-Nov-2024 (HY-2D) From 15-May-2024 to present day (Oceansat-3)
Available MY time series	From 1-Aug-1991 to 2-Jun-1996 (ERS-1) From 20-Mar-1996 to 17-Jan-2001 (ERS-2) From 19-Jul-1999 to 21-Nov-2009 (QuikSCAT) From 15-Dec-2009 to 20-Feb-2014 (Oceansat-2) From 1-Jan-2007 to 15-Nov-2021 (Metop-A) From 1-Jan-2019 to 4 months before present (Metop-B)
Dimensions	time, lat, lon
Variables	Measurement Acquisition Time [seconds since 1990-01-01 00:00:00]
	Wvc-index [no unit]
	Wvc-quality-flag [no unit]
	Backscatter Distance [n.a.]
	Air Density [kg/m^3]
	(Stress-equivalent) Wind Speed [m/s]
	Model Stress Equivalent Wind Speed [m/s]
	Wind-to Direction [degree]
	Model Wind-to Direction [degree]
	Zonal (Eastward) (Stress Equivalent) Wind Velocity [m/s]

	Model Zonal (Eastward) Stress Equivalent Wind Velocity [m/s]
	Meridional (Northward) (Stress Equivalent) Wind Velocity [m/s]
	Model Meridional (Northward) Stress Equivalent Wind Velocity [m/s]
	Wind Divergence [1/s]
	Model Stress Equivalent Wind divergence [1/s]
	Wind Curl [1/s]
	Model Stress Equivalent Wind Curl [1/s]
	Wind Stress Magnitude [N/m^2]
	Model Wind Stress Magnitude [N/m^2]
	Zonal (Eastward) Wind Stress [N/m^2]
	Zonal (Eastward) Model Wind Stress [N/m^2]
	Meridional (Northward) Wind Stress [N/m^2]
	Meridional (Northward) Model Wind Stress [N/m^2]
	Wind Stress Divergence [N/m^3]
	Model Wind Stress Divergence [N/m^3]
	Wind Stress Curl [N/m^3]
	Model Wind Stress Curl [N/m^3]

Table 1: Product Specification for WIND_GLO_PHY_L3_NRT_012_002 and WIND_GLO_PHY_L3_MY_012_005

II.1.1 L3 Global Ocean wind and stress

The global ocean L3 wind and stress product relies on L2 scatterometer ocean wind vectors which are regridded to a regular lat-lon grid with fixed spacing. The global ocean L3 wind product contains different datasets depending on satellite, grid spacing and whether the data is originating from ascending or descending passes. The latter is done to avoid temporal averaging in a basic product, as dynamical and transient changes in the wind field are much larger over a day than the required accuracy level. The L3 grid spacing matches the resolution of the input L2 product. The input L2 products are operational products from the EUMETSAT OSI SAF produced at KNMI using the EUMETSAT NWP SAF wind processors. The production of the Copernicus Marine Service L3 global ocean wind product is appended to the OSI SAF production system at KNMI (Production Centre). The resulting netCDF product files are then made available by the Copernicus Marine Service Information System.

The OSI SAF L2 netCDF input data is generated from the original L2 BUFR products and contains several additions. The model wind is converted to 10m-height stress-equivalent winds, U10S, by taking the air density into account (taken from the ECMWF NWP model). The scatterometer wind itself already is a stress equivalent wind by definition because the CMOD function used for wind retrieval reflects ocean surface roughness and no atmospheric effects. In a next step the spatial derivatives of the U10S are calculated on a half grid coordinate system defined by the swath grid and later rotated to a standard north-south oriented coordinate system for user convenience. The wind rotation and divergence and wind stress rotation and divergence are calculated on this half grid.

This methodology was chosen to enable study of error properties. Earlier versions reported only the end result of wind divergence and rotation, which contains errors from both calculation of the numerical derivatives and from interpolation from swath to regular grid. The current method splits this. All calculations of numerical derivatives are done at level 2 without any (L2 to L3) interpolation errors. Moreover, the errors caused by the L2 to L3 regridding can be studied separately.

The L3 product is derived from the L2 wind product by a gridding tool called nc_L2_to_L3 which sorts the wind vector cell measurements into cells defined by a regular lat-lon grid. The measurements from ascending and descending passes are gridded into separate datasets.The L2 to L3 regridding algorithm has been in use for long and has been tested thoroughly. For the actual wind stress calculation, the drag coefficient is taken to be linear with U10S, and has been derived by fitting a line to a drag versus U10S comparison for a full year of ERA-Interim wave model data. The resulting drag relation is very close to the well-known COARE3 relation[1]. In addition we have chosen to only include the stress equivalent wind fields. If a user needs the equivalent neutral winds, they can be derived from these using the supplied air mass density. The equations needed for this are detailed in the QUID document.

[1] J. B. Edson et al., “On the Exchange of Momentum over the Open Ocean”, 2013, DOI: 10.1175/JPO-D-12-0173.1, Journal of Physical Oceanography,43, pp.1589-1610.

The various datasets of the global ocean L3 wind and stress product in Copernicus Marine and the L2 products they are based on, are listed in Table 2. The various OSI SAF L2 wind products are visualized and fully described at https://scatterometer.knmi.nl/. As new satellite instrument data will become available within the OSI SAF, the Copernicus Marine L3 functionality will be added and the Copernicus Marine portfolio extended.

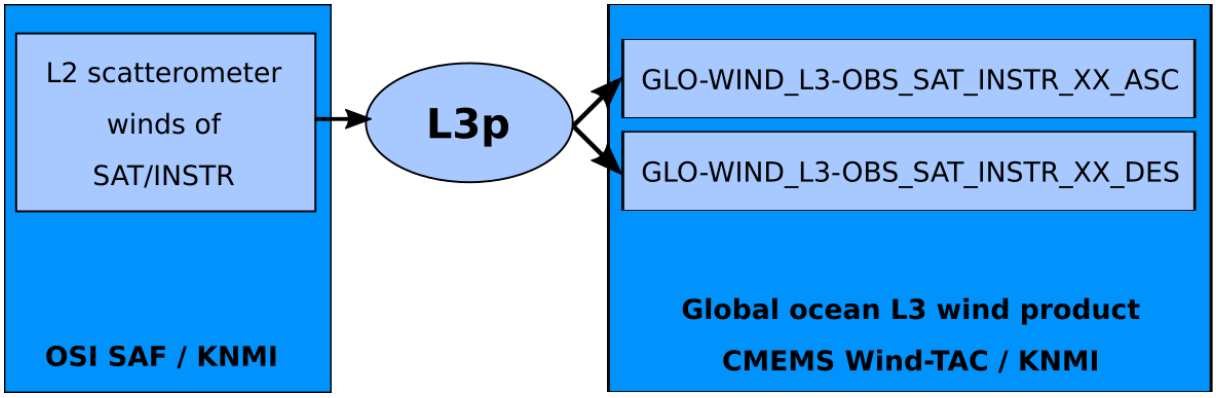

Figure 1 **Schematic view of the L2 to L3 processing (L3p) interfaces for scatterometer instrument "INSTR" on satellite "SAT".**

II.2 Details of datasets

The following tables give details on the datasets that are available in the WIND_GLO_PHY_L3_NRT_012_002 and WIND_GLO_PHY_L3_NRT_012_005 products.

Table 2 gives the L3 dataset names that are produced for the different scatterometer instruments and the corresponding L2 OSI SAF dataset identifier.

Table 3 defines the variable names and units, their corresponding netCDF long_names and CF standard names.

Table 2: List of datasets

Satellite/ Instrument	Processing type	Datasets	L2 OSI SAF Identifier
Metop-A/ASCAT	NRT	cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.25deg_P1D-i	OSI-102
		cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-des-0.25deg_P1D-i	OSI-102
		cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.125deg_P1D-i	OSI-104
		cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-des-0.125deg_P1D-i	OSI-104

Metop-A/ASCAT	MY	cmems_obs-wind_glo_phy_my_l3-metopa-ascat-asc-0.25deg_P1D-i	OSI-150-c
		cmems_obs-wind_glo_phy_my_l3-metopa-ascat-des-0.25deg_P1D-i	OSI-150-c
		cmems_obs-wind_glo_phy_my_l3-metopa-ascat-asc-0.125deg_P1D-i	OSI-150-d
		cmems_obs-wind_glo_phy_my_l3-metopa-ascat-des-0.125deg_P1D-i	OSI-150-d
Metop-B/ASCAT	NRT	cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-asc-0.25deg_P1D-i	OSI-102-b
		cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-des-0.25deg_P1D-i	OSI-102-b
		cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.125deg_P1D-i	OSI-104-b
		cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-des-0.125deg_P1D-i	OSI-104-b
Metop-B/ASCAT	MY	cmems_obs-wind_glo_phy_my_l3-metopb-ascat-asc-0.25deg_P1D-i	OSI-102-b
		cmems_obs-wind_glo_phy_my_l3-metopb-ascat-des-0.25deg_P1D-i	OSI-102-b
		cmems_obs-wind_glo_phy_my_l3-metopb-ascat-asc-0.125deg_P1D-i	OSI-104-b
		cmems_obs-wind_glo_phy_my_l3-metopb-ascat-des-0.125deg_P1D-i	OSI-104-b
Metop-C/ASCAT	NRT	cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-asc-0.25deg_P1D-i	OSI-102-c
		cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-des-0.25deg_P1D-i	OSI-102-c
		cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-asc-0.125deg_P1D-i	OSI-104-c
		cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-des-0.125deg_P1D-i	OSI-104-c
ScatSat-1/OSCAT	NRT	cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-asc-0.5deg_P1D-i	OSI-112-b
		cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-des-0.5deg_P1D-i	OSI-112-b
		cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-asc-0.25deg_P1D-i	OSI-112-a
		cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-des-0.25deg_P1D-i	OSI-112-a
Oceansat-3/OSCAT	NRT	cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-asc-0.5deg_P1D-i	OSI-113-b
		cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-des-0.5deg_P1D-i	OSI-113-b
		cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-asc-0.25deg_P1D-i	OSI-113-a
		cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-des-0.25deg_P1D-i	OSI-113-a
HY-2B/HSCAT	NRT	cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-asc-0.5deg_P1D-i	OSI-114-b
		cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-des-0.5deg_P1D-i	OSI-114-b
		cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-asc-0.25deg_P1D-i	OSI-114-a

		cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-des-0.25deg_P1D-i	OSI-114-a
HY-2C/HSCAT	NRT	cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-asc-0.5deg_P1D-i	OSI-115-b
		cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-des-0.5deg_P1D-i	OSI-115-b
		cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-des-0.25deg_P1D-i	OSI-115-a
		cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-des-0.25deg_P1D-i	OSI-115-a
HY-2D/HSCAT	NRT	cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-asc-0.5deg_P1D-i	OSI-116-b
		cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-des-0.5deg_P1D-i	OSI-116-b
		cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-asc-0.25deg_P1D-i	OSI-116-a
		cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-des-0.25deg_P1D-i	OSI-116-a
Oceansat-2/OSCAT	MY	cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-asc-0.5deg_P1D-i	OSI-153-b
		cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-des-0.5deg_P1D-i	OSI-153-b
		cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-asc-0.25deg_P1D-i	OSI-153-a
		cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-des-0.25deg_P1D-i	OSI-153-a
QuikSCAT/SeaWinds	MY	cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-asc-0.5deg_P1D-i	OSI-151-b
		cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-des-0.5deg_P1D-i	OSI-151-b
		cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-asc-0.25deg_P1D-i	OSI-151-a
		cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-des-0.25deg_P1D-i	OSI-151-a
ERS-1/SCAT	MY	cmems_obs-wind_glo_phy_my_l3-ers1-scat-asc-0.25deg_P1D-i	OSI-152
		cmems_obs-wind_glo_phy_my_l3-ers1-scat-des-0.25deg_P1D-i	OSI-152
ERS-2/SCAT	MY	cmems_obs-wind_glo_phy_my_l3-ers2-scat-asc-0.25deg_P1D-i	OSI-152
		cmems_obs-wind_glo_phy_my_l3-ers2-scat-des-0.25deg_P1D-i	OSI-152

Table 3: List of the variables and units for each dataset (first line), their long_names in the netCDF files (second line) and their CF standard names (third line) for the WIND_GLO_PHY_L3_NRT_012_002 product and the WIND_GLO_PHY_L3_MY_012_005 product

WIND_GLO_PHY_L3_NRT_012_002 and WIND_GLO_PHY_L3_MY_012_005
contain the variables
measurement_time **[seconds since 1990-01-01 00:00:00]** **measurement acquisition time** **time**
wvc_index **[no unit]** **cross track wind vector cell number** **across_swath_cell_index (proposed new name)**
wvc_quality_flag **[no unit]** **wind vector cell quality** **status_flag**
bs_distance **[n.a.]** **backscatter distance** **backscatter_distance_to_modelfunction (proposed new name)**
air_density **[kg m-3]** **air density at 10 m** **air_density**
wind_speed **[m s-1]** **stress equivalent wind speed at 10 m** **wind_speed**
se_model_speed **[m s-1]** **stress equivalent model wind speed at 10 m** **wind_speed**
wind_to_dir **[degree]** **wind direction at 10 m** **wind_to_direction**
model_wind_to_dir **[degree]** **model wind direction at 10 m** **wind_to_direction**
eastward_wind **[m s-1]** **stress equivalent wind u component at 10 m** **eastward_wind**
se_eastward_model_wind **[m s-1]** **stress equivalent model wind u component at 10 m** **eastward_wind**
northward_wind **[m s-1]** **stress equivalent wind v component at 10 m** **northward_wind**
se_northward_model_wind **[m s-1]** **stress equivalent model wind v component at 10 m** **northward_wind**

wind_divergence **[s-1]** **divergence of stress equivalent wind at 10 m** **divergence_of_wind**
se_model_wind_divergence **[s-1]** **model divergence of stress equivalent wind at 10 m** **divergence_of_wind**
wind_curl **[s-1]** **rotation of stress equivalent wind at 10 m** **atmosphere_relative_vorticity**
se_model_wind_curl **[s-1]** **model rotation of stress equivalent wind at 10 m** **atmosphere_relative_vorticity**
wind_stress_magnitude **[N m-2]** **wind stress** **magnitude_of_surface_downward_stress**
model_stress_magnitude **[N m-2]** **model stress** **magnitude_of_surface_downward_stress**
eastward_stress **[N m-2]** **wind stress u component** **surface_downward_eastward_stress**
eastward_model_stress **[N m-2]** **model stress u component** **surface_downward_eastward_stress**
northward_stress **[N m-2]** **wind stress v component** **surface_downward_northward_stress**
northward_model_stress **[N m-2]** **model stress v component** **surface_downward_northward_stress**
stress_divergence **[N m-3]** **divergence of ocean surface stress** **divergence_of_surface_downward_stress (proposed new name)**
model_stress_divergence **[N m-3]** **model divergence of ocean surface stress** **divergence_of_surface_downward_stress (proposed new name)**
stress_curl **[N m-3]** **rotation of ocean surface stress** **vertical_component_of_surface_downward_stress_curl (proposed new name)**
model_stress_curl **[N m-3]** **model rotation of ocean surface stress** **vertical_component_of_surface_downward_stress_curl (proposed new name)**

II.3 Grid

The grid that is used is a regular lat-lon grid covering the whole earth.

The grid spacing used is:

- 0.125 degrees for the gridded 12.5 km L3 scatterometer products,
- 0.25 degrees for the gridded 25 km L3 scatterometer products
- and 0.5 degrees for the gridded 50 km L3 scatterometer products.
- For the 12.5 km datasets the grid centre points go from 89.9375 degrees South to 89.9375 degrees North latitude and from 0.0625 degrees East to 359.9375 degrees East longitude.
- For the 25 km datasets the grid centre points go from 89.875 degrees South to 89.875 North latitude and from 0.125 degrees East to 359.875 degrees East longitude.
- For the 50 km datasets the grid centre points go from 89.75 degrees South to 89.75 North latitude and from 0.25 degrees East to 359.75 degrees East.

II.4 Domain coverage

The L3 global wind product consists of sea surface winds from scatterometer that are interpolated to a regular lat-lon grid. Consequently the L3 wind covers the global ocean and a equidistant cylindrical projection is used with a constant longitude and latitude step of 0.25 degrees, resp. 0.125 degrees or 0.5 degrees.

Regular projection : longitude and latitude step is constant	

II.5 Update Time

WIND_GLO_PHY_L3_NRT_012_002 product: the gridded sea surface wind observations are updated daily at around 07:00 UTC (Metop-B and Metop-C ASCAT) and 20:00 UTC (Oceansat-3 OSCAT, HY-2B, HY-2C and HY-2D HSCAT) with gridded observations of the day before.

WIND_GLO_PHY_L3_MY_012_005 product: the gridded sea surface wind observations are generated off-line when a multi-year L2 reprocessing dataset becomes available and will be added at the next Copernicus Marine Service release. The datasets cmems_obs-wind_glo_phy_my_l3-metopb-ascat-XXX-XXXdeg_P1D-i are extended in time every month, such that these datasets cover the period from 2019 up to three months before present. As L2 datasets reprocessed with ERA5 model winds are not yet available, the L2 NRT datasets are used as input from 2019 onwards.

II.6 Other information: mean centre of Products, land mask value, missing value

The L3 global wind products WIND_GLO_PHY_L3_NRT_012_002 and WIND_GLO_PHY_L3_MY_012_005 only contain observations at sea surface in the measurement swath of the satellite. The variable values for grid cells that are on land or outside the measurement swath are filled with the attribute missing_value or _FillValue. Also grid cells that contain only L3 measurements which are rejected by the OSI SAF KNMI quality control in the L2 processing (KNMI QC flag set), are filled with the attribute missing_value or _FillValue (see section IV.2 below). For more information about the quality control in the OSI SAF L2 processing see the scatterometer product handbooks .

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV NOMENCLATURE OF FILES

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1 Nomenclature of original files

WIND_GLO_PHY_L3_NRT_012_002 and WIND_GLO_PHY_L3_MY_012_005 files nomenclature when downloaded through the Copernicus Marine Web Portal is based on product dataset name and a numerical reference related to the request date on the CIS.

The scheme is: **datasetname_extractionID.nc**

where :

· **datasetname** is a character string within one of the following :

- `cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.25deg_P1D-i`
- `cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-des-0.25deg_P1D-i`
- `cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.125deg_P1D-i`
- `cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-des-0.125deg_P1D-i`
- `cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-asc-0.25deg_P1D-i`
- `cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-des-0.25deg_P1D-i`
- `cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-asc-0.125deg_P1D-i`
- `cmems_obs-wind_glo_phy_nrt_l3-metopb-ascat-des-0.125deg_P1D-i`
- `cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-asc-0.25deg_P1D-i`
- `cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-des-0.25deg_P1D-i`

- cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-asc-0.125deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-metopc-ascat-des-0.125deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-asc-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-scatsat1-oscat-des-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-asc-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-oceansat3-oscat-des-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-asc-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2b-hscat-des-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-asc-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2c-hscat-des-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-asc-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_nrt_l3-hy2d-hscat-des-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-metopa-ascat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-metopa-ascat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-metopa-ascat-asc-0.125deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-metopa-ascat-des-0.125deg_P1D-i

- cmems_obs-wind_glo_phy_my_l3-metopb-ascat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-metopb-ascat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-metopb-ascat-asc-0.125deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-metopb-ascat-des-0.125deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-asc-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-quikscat-seawinds-des-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-ers1-scat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-ers1-scat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-ers2-scat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-ers2-scat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-asc-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-des-0.25deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-asc-0.5deg_P1D-i
- cmems_obs-wind_glo_phy_my_l3-oceansat2-oscat-des-0.5deg_P1D-i

Where the name indicates the satellite (Metop-A, Metop-B, Metop-C, Oceansat-2, Oceansat-3, ScatSat-1, HY-2B, HY-2C, HY-2D, QuikSCAT or ERS-1/ERS-2), the instrument (ASCAT, OSCAT, SeaWinds, HSCAT or SCAT), the horizontal grid resolution (0.125°, 0.25° or 0.5°), and whether the data are coming from ascending or descending passes.

· **extractionID**: 13 digit integer corresponding to the extraction (subsetting) operation (uniquely identified)

.nc: standard netCDF filename extension.

Example:

cmems_obs-wind_glo_phy_nrt_l3-metopa-ascat-asc-0.125deg_P1D-i_1468420761203.nc

IV.2 Structure of files

As already mentioned in the introduction of this document, significant changes have been implemented as dataset version V2. This format is applicable for the current NRT product based on the ASCAT on Metop-A/B/C data sets and all REP datasets, and it will be used from now on for all new near real time and reprocessed datasets as well. The V2 file format used by all other datasets is detailed next (as generated by “ncdump –h”). An Example of the header of output NetCDF files is inserted in **annex**.

V REFERENCE

Quality Information Document

CMEMS-ARC-QUID-002-005.pdf

VI ANNEX

```
netcdf GLO-WIND_L3-OBS_METOP-A_ASCAT_25_ASC_20160710 {
dimensions:
        time = 1 ;
        lat = 720 ;
        lon = 1440 ;
variables:
int time(time) ;
                time:standard_name = "time" ;
                time:long_name = "Validity time" ;
                time:units = "seconds since 1990-01-01 00:00:00" ;
                time:calendar = "Gregorian" ;
                time:axis = "T" ;
        float lat(lat) ;
                lat:valid_min = -90.f ;
                lat:valid_max = 90.f ;
                lat:standard_name = "latitude" ;
                lat:long_name = "latitude" ;
                lat:units = "degrees_north" ;
                lat:axis = "Y" ;
        float lon(lon) ;
                lon:valid_min = 0.f ;
                lon:valid_max = 360.f ;
                lon:standard_name = "longitude" ;
                lon:long_name = "longitude" ;
                lon:units = "degrees_east" ;
                lon:axis = "X" ;
        int measurement_time(time, lat, lon) ;
                measurement_time:_FillValue = -2147483647 ;
                measurement_time:missing_value = -2147483647 ;
                measurement_time:valid_min = 0 ;
                measurement_time:valid_max = 2147483647 ;
                measurement_time:standard_name = "time" ;
```

```
            measurement_time:long_name = "measurement acquisition time" ;
            measurement_time:units = "seconds since 1990-01-01 00:00:00" ;
            measurement_time:coordinates = "time lat lon" ;
    short wvc_index(time, lat, lon) ;
            wvc_index:_FillValue = -32767s ;
            wvc_index:missing_value = -32767s ;
            wvc_index:valid_min = 0s ;
            wvc_index:valid_max = 999s ;
            wvc_index:standard_name = "across_swath_cell_index" ;
            wvc_index:long_name = "cross track wind vector cell number" ;
            wvc_index:units = "1" ;
            wvc_index:coordinates = "time lat lon" ;
    short air_density(time, lat, lon) ;
            air_density:_FillValue = -32767s ;
            air_density:missing_value = -32767s ;
            air_density:valid_min = 0s ;
            air_density:valid_max = 2000s ;
            air_density:standard_name = "air_density" ;
            air_density:long_name = "air density at 10 m" ;
            air_density:units = "kg m-3" ;
            air_density:scale_factor = 0.001 ;
            air_density:add_offset = 0. ;
            air_density:coordinates = "time lat lon" ;
            air_density:air_density_source = "ECMWF (Operational Model)" ;
    int stress_divergence(time, lat, lon) ;
            stress_divergence:_FillValue = -2147483647 ;
            stress_divergence:missing_value = -2147483647 ;
            stress_divergence:valid_min = -500000000 ;
            stress_divergence:valid_max = 500000000 ;
            stress_divergence:standard_name = "divergence_of_surface_downward_stress" ;
            stress_divergence:long_name = "divergence of ocean surface stress" ;
            stress_divergence:units = "N m-3" ;
            stress_divergence:scale_factor = 1.e-10 ;
            stress_divergence:add_offset = 0. ;
            stress_divergence:coordinates = "time lat lon" ;
```

```
int stress_curl(time, lat, lon) ;
        stress_curl:_FillValue = -2147483647 ;
        stress_curl:missing_value = -2147483647 ;
        stress_curl:valid_min = -500000000 ;
        stress_curl:valid_max = 500000000 ;
        stress_curl:standard_name = "vertical_component_of_surface_downward_stress_curl" ;
        stress_curl:long_name = "rotation of ocean surface stress" ;
        stress_curl:units = "N m-3" ;
        stress_curl:scale_factor = 1.e-10 ;
        stress_curl:add_offset = 0. ;
        stress_curl:coordinates = "time lat lon" ;
int model_stress_divergence(time, lat, lon) ;
        model_stress_divergence:_FillValue = -2147483647 ;
        model_stress_divergence:missing_value = -2147483647 ;
        model_stress_divergence:valid_min = -500000000 ;
        model_stress_divergence:valid_max = 500000000 ;
        model_stress_divergence:standard_name = "divergence_of_surface_downward_stress" ;
        model_stress_divergence:long_name = "model divergence of ocean surface stress" ;
        model_stress_divergence:units = "N m-3" ;
        model_stress_divergence:scale_factor = 1.e-10 ;
        model_stress_divergence:add_offset = 0. ;
        model_stress_divergence:coordinates = "time lat lon" ;
        model_stress_divergence:background_wind_source = "ECMWF (Operational Model)" ;
int model_stress_curl(time, lat, lon) ;
        model_stress_curl:_FillValue = -2147483647 ;
        model_stress_curl:missing_value = -2147483647 ;
        model_stress_curl:valid_min = -500000000 ;
        model_stress_curl:valid_max = 500000000 ;
        model_stress_curl:standard_name = "vertical_component_of_surface_downward_stress_curl"
;
        model_stress_curl:long_name = "model rotation of ocean surface stress" ;
        model_stress_curl:units = "N m-3" ;
        model_stress_curl:scale_factor = 1.e-10 ;
        model_stress_curl:add_offset = 0. ;
        model_stress_curl:coordinates = "time lat lon" ;
```

```
            model_stress_curl:background_wind_source = "ECMWF (Operational Model)" ;
        int wind_divergence(time, lat, lon) ;
            wind_divergence:_FillValue = -2147483647 ;
            wind_divergence:missing_value = -2147483647 ;
            wind_divergence:valid_min = -500000 ;
            wind_divergence:valid_max = 500000 ;
            wind_divergence:standard_name = "divergence_of_wind" ;
            wind_divergence:long_name = "divergence of stress equivalent wind at 10m" ;
            wind_divergence:units = "s-1" ;
            wind_divergence:scale_factor = 1.e-07 ;
            wind_divergence:add_offset = 0. ;
            wind_divergence:coordinates = "time lat lon" ;
        int wind_curl(time, lat, lon) ;
            wind_curl:_FillValue = -2147483647 ;
            wind_curl:missing_value = -2147483647 ;
            wind_curl:valid_min = -500000 ;
            wind_curl:valid_max = 500000 ;
            wind_curl:standard_name = "atmosphere_relative_vorticity" ;
            wind_curl:long_name = "rotation of stress equivalent wind at 10m" ;
            wind_curl:units = "s-1" ;
            wind_curl:scale_factor =1.e-07 ;
            wind_curl:add_offset = 0. ;
            wind_curl:coordinates = "time lat lon" ;
        int se_model_wind_divergence(time, lat, lon) ;
            se_model_wind_divergence:_FillValue = -2147483647 ;
            se_model_wind_divergence:missing_value = -2147483647 ;
            se_model_wind_divergence:valid_min = -500000 ;
            se_model_wind_divergence:valid_max = 500000 ;
            se_model_wind_divergence:standard_name = "divergence_of_wind" ;
            se_model_wind_divergence:long_name = "model divergence of stress equivalent wind at 10m"
;
            se_model_wind_divergence:units = "s-1" ;
            se_model_wind_divergence:scale_factor = 1.e-07 ;
            se_model_wind_divergence:add_offset = 0. ;
            se_model_wind_divergence:coordinates = "time lat lon" ;
```

```
        se_model_wind_divergence:background_wind_source = "ECMWF (Operational Model)" ;
int se_model_wind_curl(time, lat, lon) ;
        se_model_wind_curl:_FillValue = -2147483647 ;
        se_model_wind_curl:missing_value = -2147483647 ;
        se_model_wind_curl:valid_min = -500000 ;
        se_model_wind_curl:valid_max = 500000 ;
        se_model_wind_curl:standard_name = "atmosphere_relative_vorticity" ;
        se_model_wind_curl:long_name = "model rotation of stress equivalent wind at 10m" ;
        se_model_wind_curl:units = "s-1" ;
        se_model_wind_curl:scale_factor = 1.e-07 ;
        se_model_wind_curl:add_offset = 0. ;
        se_model_wind_curl:coordinates = "time lat lon" ;
        se_model_wind_curl:background_wind_source = "ECMWF (Operational Model)" ;
short se_model_speed(time, lat, lon) ;
        se_model_speed:_FillValue = -32767s ;
        se_model_speed:missing_value = -32767s ;
        se_model_speed:valid_min = 0s ;
        se_model_speed:valid_max = 5000s ;
        se_model_speed:standard_name = "wind_speed" ;
        se_model_speed:long_name = "stress equivalent model wind speed at 10 m" ;
        se_model_speed:units = "m s-1" ;
        se_model_speed:scale_factor = 0.01 ;
        se_model_speed:add_offset = 0. ;
        se_model_speed:coordinates = "time lat lon" ;
        se_model_speed:background_wind_source = "ECMWF (Operational Model)" ;
short model_wind_to_dir(time, lat, lon) ;
        model_wind_to_dir:_FillValue = -32767s ;
        model_wind_to_dir:missing_value = -32767s ;
        model_wind_to_dir:valid_min = 0s ;
        model_wind_to_dir:valid_max = 3600s ;
        model_wind_to_dir:standard_name = "wind_to_direction" ;
        model_wind_to_dir:long_name = "model wind direction at 10 m" ;
        model_wind_to_dir:units = "degree" ;
        model_wind_to_dir:scale_factor = 0.1 ;
        model_wind_to_dir:add_offset = 0. ;
```

```
                model_wind_to_dir:coordinates = "time lat lon" ;
                model_wind_to_dir:background_wind_source = "ECMWF (Operational Model)" ;
        int wvc_quality_flag(time, lat, lon) ;
                wvc_quality_flag:_FillValue = -2147483647 ;
                wvc_quality_flag:missing_value = -2147483647 ;
                wvc_quality_flag:valid_min = 0 ;
                wvc_quality_flag:valid_max = 8388607 ;
                wvc_quality_flag:standard_name = "status_flag" ;
                wvc_quality_flag:long_name = "wind vector cell quality" ;
                wvc_quality_flag:coordinates = "time lat lon" ;
                wvc_quality_flag:flag_masks = 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, 65536,
131072, 262144, 524288, 1048576, 2097152, 4194304 ;
                wvc_quality_flag:flag_meanings = "distance_to_gmf_too_large data_are_redundant
no_meteorological_background_used rain_detected rain_flag_not_usable
small_wind_less_than_or_equal_to_3_m_s large_wind_greater_than_30_m_s wind_inversion_not_successful
some_portion_of_wvc_is_over_ice some_portion_of_wvc_is_over_land variational_quality_control_fails
knmi_quality_control_fails product_monitoring_event_flag product_monitoring_not_used
any_beam_noise_content_above_threshold poor_azimuth_diversity
not_enough_good_sigma0_for_wind_retrieval" ;
        short wind_speed(time, lat, lon) ;
                wind_speed:_FillValue = -32767s ;
                wind_speed:missing_value = -32767s ;
                wind_speed:valid_min = 0s ;
                wind_speed:valid_max = 5000s ;
                wind_speed:standard_name = "wind_speed" ;
                wind_speed:long_name = "stress equivalent wind speed at 10 m" ;
                wind_speed:units = "m s-1" ;
                wind_speed:scale_factor = 0.01 ;
                wind_speed:add_offset = 0. ;
                wind_speed:coordinates = "time lat lon" ;
        short wind_to_dir(time, lat, lon) ;
                wind_to_dir:_FillValue = -32767s ;
                wind_to_dir:missing_value = -32767s ;
                wind_to_dir:valid_min = 0s ;
                wind_to_dir:valid_max = 3600s ;
                wind_to_dir:standard_name = "wind_to_direction" ;
                wind_to_dir:long_name = "wind direction at 10 m" ;
```

```
        wind_to_dir:units = "degree" ;
        wind_to_dir:scale_factor = 0.1 ;
        wind_to_dir:add_offset = 0. ;
        wind_to_dir:coordinates = "time lat lon" ;
    short eastward_wind(time, lat, lon) ;
        eastward_wind:_FillValue = -32767s ;
        eastward_wind:missing_value = -32767s ;
        eastward_wind:valid_min = -5000s ;
        eastward_wind:valid_max = 5000s ;
        eastward_wind:standard_name = "eastward_wind" ;
        eastward_wind:long_name = "stress equivalent wind u component at 10 m" ;
        eastward_wind:units = "m s-1" ;
        eastward_wind:scale_factor = 0.01 ;
        eastward_wind:add_offset = 0. ;
        eastward_wind:coordinates = "time lat lon" ;
    short northward_wind(time, lat, lon) ;
        northward_wind:_FillValue = -32767s ;
        northward_wind:missing_value = -32767s ;
        northward_wind:valid_min = -5000s ;
        northward_wind:valid_max = 5000s ;
        northward_wind:standard_name = "northward_wind" ;
        northward_wind:long_name = "stress equivalent wind v component at 10 m" ;
        northward_wind:units = "m s-1" ;
        northward_wind:scale_factor = 0.01 ;
        northward_wind:add_offset = 0. ;
        northward_wind:coordinates = "time lat lon" ;
    short se_eastward_model_wind(time, lat, lon) ;
        se_eastward_model_wind:_FillValue = -32767s ;
        se_eastward_model_wind:missing_value = -32767s ;
        se_eastward_model_wind:valid_min = -5000s ;
        se_eastward_model_wind:valid_max = 5000s ;
        se_eastward_model_wind:standard_name = "eastward_wind" ;
        se_eastward_model_wind:long_name = "stress equivalent model wind u component at 10 m" ;
        se_eastward_model_wind:units = "m s-1" ;
        se_eastward_model_wind:scale_factor = 0.01 ;
```

```
        se_eastward_model_wind:add_offset = 0. ;
        se_eastward_model_wind:coordinates = "time lat lon" ;
        se_eastward_model_wind:background_wind_source = "ECMWF (Operational Model)" ;
    short se_northward_model_wind(time, lat, lon) ;
        se_northward_model_wind:_FillValue = -32767s ;
        se_northward_model_wind:missing_value = -32767s ;
        se_northward_model_wind:valid_min = -5000s ;
        se_northward_model_wind:valid_max = 5000s ;
        se_northward_model_wind:standard_name = "northward_wind" ;
        se_northward_model_wind:long_name = "stress equivalent model wind v component at 10 m"
;
        se_northward_model_wind:units = "m s-1" ;
        se_northward_model_wind:scale_factor = 0.01 ;
        se_northward_model_wind:add_offset = 0. ;
        se_northward_model_wind:coordinates = "time lat lon" ;
        se_northward_model_wind:background_wind_source = "ECMWF (Operational Model)" ;
    int wind_stress_magnitude(time, lat, lon) ;
        wind_stress_magnitude:_FillValue = -2147483647 ;
        wind_stress_magnitude:missing_value = -2147483647 ;
        wind_stress_magnitude:valid_min = 0 ;
        wind_stress_magnitude:valid_max = 5000 ;
        wind_stress_magnitude:standard_name = "magnitude_of_surface_downward_stress" ;
        wind_stress_magnitude:long_name = "wind stress" ;
        wind_stress_magnitude:units = "N m-2" ;
        wind_stress_magnitude:scale_factor = 0.01 ;
        wind_stress_magnitude:add_offset = 0. ;
        wind_stress_magnitude:coordinates = "time lat lon" ;
    int model_stress_magnitude(time, lat, lon) ;
        model_stress_magnitude:_FillValue = -2147483647 ;
        model_stress_magnitude:missing_value = -2147483647 ;
        model_stress_magnitude:valid_min = 0 ;
        model_stress_magnitude:valid_max = 5000 ;
        model_stress_magnitude:standard_name = "magnitude_of_surface_downward_stress" ;
        model_stress_magnitude:long_name = "model stress" ;
        model_stress_magnitude:units = "N m-2" ;
```

```
        model_stress_magnitude:scale_factor = 0.01 ;
        model_stress_magnitude:add_offset = 0. ;
        model_stress_magnitude:coordinates = "time lat lon" ;
        model_stress_magnitude:background_wind_source = "ECMWF (Operational Model)" ;
int eastward_stress(time, lat, lon) ;
        eastward_stress:_FillValue = -2147483647 ;
        eastward_stress:missing_value = -2147483647 ;
        eastward_stress:valid_min = -5000 ;
        eastward_stress:valid_max = 5000 ;
        eastward_stress:standard_name = "surface_downward_eastward_stress" ;
        eastward_stress:long_name = "wind stress u component" ;
        eastward_stress:units = "N m-2" ;
        eastward_stress:scale_factor = 0.01 ;
        eastward_stress:add_offset = 0. ;
        eastward_stress:coordinates = "time lat lon" ;
int northward_stress(time, lat, lon) ;
        northward_stress:_FillValue = -2147483647 ;
        northward_stress:missing_value = -2147483647 ;
        northward_stress:valid_min = -5000 ;
        northward_stress:valid_max = 5000 ;
        northward_stress:standard_name = "surface_downward_northward_stress" ;
        northward_stress:long_name = "wind stress v component" ;
        northward_stress:units = "N m-2" ;
        northward_stress:scale_factor = 0.01 ;
        northward_stress:add_offset = 0. ;
        northward_stress:coordinates = "time lat lon" ;
int eastward_model_stress(time, lat, lon) ;
        eastward_model_stress:_FillValue = -2147483647 ;
        eastward_model_stress:missing_value = -2147483647 ;
        eastward_model_stress:valid_min = -5000 ;
        eastward_model_stress:valid_max = 5000 ;
        eastward_model_stress:standard_name = "surface_downward_eastward_stress" ;
        eastward_model_stress:long_name = "model stress u component" ;
        eastward_model_stress:units = "N m-2" ;
        eastward_model_stress:scale_factor = 0.01 ;
```

```
            eastward_model_stress:add_offset = 0. ;
            eastward_model_stress:coordinates = "time lat lon" ;
            eastward_model_stress:background_wind_source = "ECMWF (Operational Model)" ;
        int northward_model_stress(time, lat, lon) ;
            northward_model_stress:_FillValue = -2147483647 ;
            northward_model_stress:missing_value = -2147483647 ;
            northward_model_stress:valid_min = -5000 ;
            northward_model_stress:valid_max = 5000 ;
            northward_model_stress:standard_name = "surface_downward_northward_stress" ;
            northward_model_stress:long_name = "model stress v component" ;
            northward_model_stress:units = "N m-2" ;
            northward_model_stress:scale_factor = 0.01 ;
            northward_model_stress:add_offset = 0. ;
            northward_model_stress:coordinates = "time lat lon" ;
            northward_model_stress:background_wind_source = "ECMWF (Operational Model)" ;
        short bs_distance(time, lat, lon) ;
            bs_distance:_FillValue = -32767s ;
            bs_distance:missing_value = -32767s ;
            bs_distance:valid_min = -500s ;
            bs_distance:valid_max = 500s ;
            bs_distance:standard_name = "backscatter_distance_to_modelfunction" ;
            bs_distance:long_name = "backscatter distance" ;
            bs_distance:units = "1" ;
            bs_distance:scale_factor = 0.1 ;
            bs_distance:add_offset = 0. ;
            bs_distance:coordinates = "time lat lon" ;

// global attributes:
            :title = " Global Ocean - Wind - METOP-A ASCAT - 25km daily Ascending V2" ;
            :title_short_name = "ASCATA-L3-25km" ;
            :Conventions = "CF-1.6" ;
            :institution = "EUMETSAT/OSI SAF/KNMI" ;
            :source = "MetOp-A ASCAT" ;
            :software_identification_level_1 = 1000 ;
            :instrument_calibration_version = 0 ;
```

:software_identification_wind = 2401 ;

:pixel_size_on_horizontal = "25.0 km" ;

:service_type = "N/A" ;

:processing_type = "O" ;

:contents = "ovw" ;

:granule_name = "GLO-WIND_L3-OBS_METOP-A_ASCAT_25_ASC_20160710.nc" ;

:processing_level = "L3" ;

:orbit_number = 50453 ;

:start_date = "2016-07-10" ;

:start_time = "00:00:00" ;

:stop_date = "2016-07-10" ;

:stop_time = "23:59:56" ;

:equator_crossing_longitude = " 328.246" ;

:equator_crossing_date = "2016-07-09" ;

:equator_crossing_time = "23:35:50" ;

:rev_orbit_period = "6081.7" ;

:orbit_inclination = "98.7" ;

:history = "N/A" ;

:references = "ASCAT Wind Product User Manual, http://www.osi-saf.org/, http://www.knmi.nl/scatterometer/" ;

:comment = "Orbit period and inclination are constant values. All wind directions in oceanographic convention (0 deg. flowing North)" ;

:creation_date = "2016-07-11" ;

:creation_time = "08:11:18" ;

}